Chapter III.6, Week 1

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6.1. Let $(X, \mathscr{O}_X)$ be a ringed space, and let $\mathscr{F}', \mathscr{F}'' \in \mathfrak{Mod}(X)$. An extension of $\mathscr{F}''$ by $\mathscr{F}'$ is a short exact sequence

$$0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$$

in $\mathfrak{Mod}(X)$. Two extensions are *isomorphic* if there is an isomorphism of the short exact seuqences, inducing the identity maps on $\mathscr{F}'$ and $\mathscr{F}''$. Given an extension as above, consider the long exact sequence arising from $\mathrm{Hom}(\mathscr{F}'', \cdot)$, in particular the map

$$\delta : \mathrm{Hom}(\mathscr{F}'', \mathscr{F}'') \to \mathrm{Ext}^1(\mathscr{F}'', \mathscr{F}'),$$

and let $\xi \in \mathrm{Ext}^1(\mathscr{F}'', \mathscr{F}')$ be $\delta(1_{\mathscr{F}''})$. Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of $\mathscr{F}''$ by $\mathscr{F}'$, and elements of the group $\mathrm{Ext}^1(\mathscr{F}'', \mathscr{F}')$. For more details, see, e.g., Hilton and Stammbach, Ch. III.

Proof. (1) We define a map from $\mathrm{Ext}^1(\mathscr{F}'', \mathscr{F}')$ to extensions $0 \to \mathscr{F}' \to \mathscr{E} \to \mathscr{F}'' \to 0$ as follows. Fix an injective resolution $0 \to \mathscr{F}' \xrightarrow{d^{-1}} \mathscr{I}^\bullet$ of $\mathscr{F}'$ with maps $d^i : \mathscr{I}^i \to \mathscr{I}^{i+1}$, so that

$$\begin{aligned} \mathrm{Ext}^1(\mathscr{F}'', \mathscr{F}') &\cong h^1(\mathrm{Hom}(\mathscr{F}'', \mathscr{I}^\bullet)) \\ &\cong \frac{\ker d^1}{\mathrm{im}\, d^0}. \end{aligned}$$

Thus, we can regard an element $[\xi] \in \mathrm{Ext}^1(\mathscr{F}'', \mathscr{F}')$ as the class of a map $\xi : \mathscr{F}'' \to \mathscr{I}^1$ with $d^1\xi = 0$. Fix such an element $[\xi]$ with representative ξ, and take the pullout $\mathscr{P} := \mathscr{I}^0 \times_{\mathscr{I}^1} \mathscr{F}''$ in the following diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{F}' & \xrightarrow{d^{-1}} & \mathscr{I}^0 & \xrightarrow{d^0} & \mathscr{I}^1 & \xrightarrow{d^1} & \mathscr{I}^2 \\ & & & & {\scriptstyle\gamma}\uparrow & \urcorner & \uparrow{\scriptstyle\xi} & & \\ & & & & \mathscr{P} & \xrightarrow[\psi]{} & \mathscr{F}'' & & \end{array}$$

Then we have $d^0\phi = 0$, so the universal property of $\mathscr{P}$ supplies a unique map $\phi : \mathscr{F}' \to \mathscr{P}$ such that the following diagram commutes:

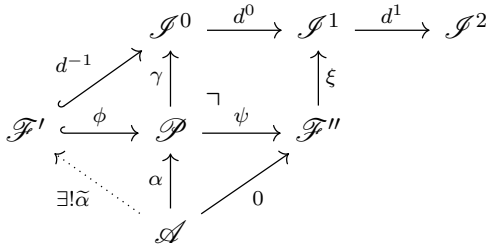
$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{F}' & \xrightarrow{d^{-1}} & \mathscr{I}^0 & \xrightarrow{d^0} & \mathscr{I}^1 & \xrightarrow{d^1} & \mathscr{I}^2 \\ & & & \searrow^{d^{-1}} & {\scriptstyle\gamma}\uparrow & & \uparrow{\scriptstyle\xi} & & \\ & & \mathscr{F}' & \xrightarrow{\exists!\phi} & \mathscr{P} & \xrightarrow[\psi]{} & \mathscr{F}'' & & \\ & & & \searrow & & \nearrow_{0} & & & \end{array}$$

We will now show that $0 \to \mathscr{F}' \xrightarrow{\phi} \mathscr{P} \xrightarrow{\psi} \mathscr{F}'' \to 0$ is a short exact sequence.

ϕ is monic since $d^{-1} = \gamma\phi$ is monic, and $\psi\phi = 0$ by construction. To show exactness at the middle, it suffices to show that $\operatorname{im}\phi$ satisfies the universal property of $\ker\psi$. Let $\alpha : \mathscr{A} \to \mathscr{P}$ be a morphism such that $\psi\alpha = 0$. It suffices to show that α factors through $\operatorname{im}\phi$. Recall that ϕ, d^{-1} are monic and $\mathscr{F}' \to \mathscr{I}^0 \to \mathscr{I}^1$ is exact, so

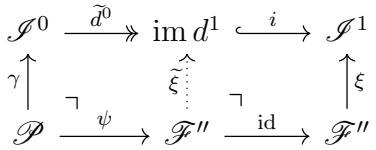
$$\operatorname{im}\phi \cong \operatorname{coim}\phi \cong \mathscr{F}' \cong \operatorname{im} d^{-1} \cong \ker d^0.$$

Then $d^0\gamma\alpha = \xi\psi\alpha = 0$, so the universal property of kernels supplies a unique map $\widetilde{\alpha} : \mathscr{A} \to \mathscr{F}'$ such that $\gamma\alpha = d^{-1}\widetilde{\alpha}$. Further, $\gamma\phi\widetilde{\alpha} = \gamma\alpha$ and $\psi\phi\widetilde{\alpha} = 0 = \psi\alpha$ imply that $\alpha = \phi\widetilde{\alpha}$ via universal property of pullbacks.



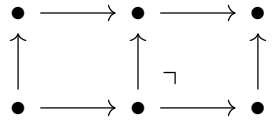
Hence, the above diagram commutes, and $\widetilde{\alpha}$ is indeed unique since ϕ is monic.

Finally, we show that ψ is epic. $d^1\xi = 0$ allows us to factor ξ through $\ker d^1 = \operatorname{im} d^0$, giving us the following diagram, where the rightmost square is a pullback.



It follows from the following elementary lemma on pullbacks that the lefthand square is a pullback as well.

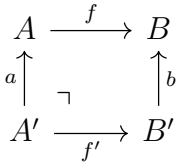
Lemma 1. *Suppose that the rightmost square is a pullback. Then the leftmost square is a pullback if and only if the entire rectangle is a pullback.*



ψ epic will follow from the following lemma.

Lemma 2. *Pullbacks in an abelian category preserve epimorphisms.*[1]

Proof. Consider the following pullback square.



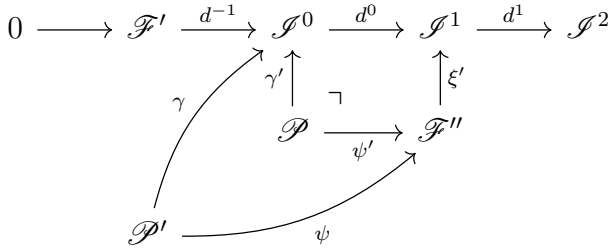
[1] I went on a day long wild goose chase trying to prove this with a more standard diagram chase using that all epimorphisms in an abelian category are cokernels. I originally thought it was a general fact that coequalizers are preserved by pullbacks, but that didn't last long, thankfully.

We are working in an abelian category, so that the following sequence is left exact.

$$0 \longrightarrow A' \xrightarrow{\binom{a}{f'}} A \oplus B' \xrightarrow{(f\ -b)} B \longrightarrow 0$$

f epic implies $(f\ -b)$ is epic, so that the sequence is right exact as well. Hence, our original square was bicartesian, and f' must be epic since pushforwards preserve epimorphisms in an arbitrary category. □

(2) Now, we must show the extension $0 \to \mathscr{F}' \xrightarrow{\phi} \mathscr{P} \xrightarrow{\psi} \mathscr{F}'' \to 0$ depends only on the cohomology class of ξ. Let $\xi, \xi' \in \mathrm{Hom}(\mathscr{F}'', \mathscr{I}^1)$ be cohomologous maps, and take $\zeta \in \mathrm{Hom}(\mathscr{F}'', \mathscr{I}^0)$ such that $\xi = \xi' + d^0\zeta$. Consider the following diagram.



We have

$$\xi'\psi = \xi\psi - d^0\zeta\psi = d^0\gamma - d^0\zeta\psi,$$

so that $d^0\gamma - \xi'\psi = d^0\zeta\psi$. We proceed by giving up.

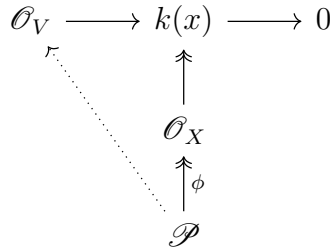
(3) It suffices to show that our two maps are indeed inverse to eachother. □

6.2. Let $X = \mathbb{P}^1_k$, with k an infinite field.

(a) Show that there does not exist a projective object $\mathscr{P} \in \mathfrak{Mod}(X)$, together with a surjective map $\mathscr{P} \to \mathscr{O}_X \to 0$. [*Hint*: Consider surjections of the form $\mathscr{O}_V \to k(x) \to 0$, where $x \in X$ is a closed point, V is an open neighborhood of x, and $\mathscr{O}_V = j_!(\mathscr{O}_X|_V)$, where $j : V \to X$ is the inclusion.]

(b) Show that there does not exist a projective object $\mathscr{P}$ in either $\mathfrak{QCoh}(X)$ or $\mathfrak{Coh}(X)$ together with a surjection $\mathscr{P} \to \mathscr{O}_X \to 0$ [*Hint*: Consider surjections of the form $\mathscr{L} \to \mathscr{L} \otimes k(x) \to 0$, where $x \in X$ is a closed point, and $\mathscr{L}$ is an invertible sheaf on X.]

Proof. (a) Suppose not, and take such a $\mathscr{P} \in \mathfrak{Mod}(X)$ and surjective map $\phi : \mathscr{P} \twoheadrightarrow \mathscr{O}_X$. Take a closed point $x \in X$, and let $U, V \ni x$ be neighborhoods of x such that $U \not\subset V$ and $V \not\subset U$. Then we have a surjection $j_!(\mathscr{O}_X|_V) =: \mathscr{O}_V \twoheadrightarrow k(x) := i_*(k(x))$, and hence the following diagram via the projective lifting property.



Now, take $s \in \Gamma(U, \mathscr{P})$. We have $\Gamma(U, \mathscr{O}_V) = 0$ since $V \not\subset U$,so $\phi(s) \mapsto 0$ under $\mathscr{O}_X \twoheadrightarrow k(x)$. But then $\mathrm{im}\,\phi_x \subset m_x \subsetneq \mathscr{O}_{X,x}$, contradicting surjectivity of ϕ.

(b) Suppose not, and take such a $\mathscr{P} \in \mathfrak{Mod}(X)$ and surjective map $\mathscr{P} \to \mathscr{O}_X \to 0$. Fix $x \in X$ and consider the surjection $\mathscr{O}_X \to k(x) \to 0$. (Ex. 5.10) gives us $n > 0$ such that both $\Gamma(X, \mathscr{O}_X(n)) \to k(x) \to 0$ and $\Gamma(X, \mathscr{P}(n)) \to \Gamma(X, \mathscr{O}_X(n)) \to 0$ are surjective. Tensoring $\mathscr{O}_X \to k(x) \to 0$ with $\mathscr{O}_X(-n-1)$ gives us a surjection

$$\mathscr{O}_X(-n-1) \to k(x) \otimes \mathscr{O}_X(-n-1) \cong k(x) \to 0$$

Once again, the projective criterion gives us a map $\mathscr{P} \to \mathscr{O}_X(-n-1)$ such that the following diagram commutes.

$$\begin{array}{ccccc} \mathscr{O}_X(-n-1) & \longrightarrow & k(x) & \longrightarrow & 0 \\ & \ulcorner \cdots & \uparrow & & \\ & & \mathscr{O}_X & & \\ & & \uparrow & & \\ & & \mathscr{P} & & \end{array}$$

We can tensor this diagram with $\mathscr{O}_X(n)$.

$$\begin{array}{ccccc} \mathscr{O}_X(-1) & \longrightarrow & k(x) & \longrightarrow & 0 \\ & \ulcorner \cdots & \uparrow & & \\ & & \mathscr{O}_X(n) & & \\ & & \uparrow & & \\ & & \mathscr{P}(n) & & \end{array}$$

Then the following diagram commutes.

$$\begin{array}{ccc} 0 = \Gamma(X, \mathscr{O}_X(-1)) & \longrightarrow & k(x) \neq 0 \\ & \nwarrow & \uparrow \\ & & \Gamma(X, \mathscr{O}_X(n)) \\ & & \uparrow \\ & & \Gamma(X, \mathscr{P}(n)) \end{array}$$

However, this implies $0 = \Gamma(X, \mathscr{O}_X(-1)) \to k(x) \neq 0$ is a surjection, which is a contradiction. □

6.3. Let X be a noetherian, and let $\mathscr{F}, \mathscr{G} \in \mathfrak{Mod}(X)$.
   (a) If $\mathscr{F}, \mathscr{G}$ are both coherent, then $\mathscr{E}xt^i(\mathscr{F}, \mathscr{G})$ is coherent, for all $i \geqslant 0$.
   (b) If $\mathscr{F}$ is coherent and $\mathscr{G}$ is quasi-coherent, then $\mathscr{E}xt^i(\mathscr{F}, \mathscr{G})$ is quasi-coherent, for all $i \geqslant 0$.

Proof. (a) We begin by showing the result for $i = 0$. Take an affine open $\operatorname{Spec} A \cong U \subset X$ and finitely generated A-modules M, N such that $\mathscr{F}|_U \cong \widetilde{M}$ and $\mathscr{G}|_U \cong \widetilde{N}$. Then we have

$$\operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U) \cong \operatorname{Hom}_A(M, N).$$

We claim that $\mathrm{Hom}_A(M,N)$ is a finitely generated A-module. Take presentations $A^l \to A^m \to M \to 0$ and $A^l \to A^n \to N \to 0$, so that the following diagram commutes.

$$\begin{array}{ccc} A^{m\times n} \cong \mathrm{Hom}(A^m, A^n) & \twoheadrightarrow & \mathrm{Hom}(A^m, N) \\ & & \uparrow \\ & & \mathrm{Hom}(M,N) \end{array}$$

Then $\mathrm{Hom}(M,N)$ is a submodule of a finitely generated module, and thus finitely generated since A is noetherian.

In particular, we can take a distinguished affine $D(f) \subset \operatorname{Spec} A$ for $f \in A$, and see via (6.2) that

$$\mathrm{Hom}_{\mathscr{O}_{D(f)}}(\mathscr{F}|_{D(f)}), \mathscr{G}|_{D(f)}) \cong \mathrm{Hom}_{A_f}(M_f, N_f).$$

We claim that

$$\mathrm{Hom}_{A_f}(M_f, N_f) \cong \mathrm{Hom}_A(M,N)_f,$$

whence it would follow from the above that

$$\mathcal{H}om_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U) \cong \widetilde{\mathrm{Hom}_A(M,N)}.$$

Fix a presentations $A^l \to A^m \to M \to 0$. Clearly the result holds when M is a free module, as

$$\mathrm{Hom}_A(A^m, N)_f \cong N_f^{\oplus m} \cong \mathrm{Hom}_{A_f}(A_f^m, N).$$

Localization is exact, so the above gives us that the following sequence is exact,

$$0 \longrightarrow \mathrm{Hom}_A(M,N)_f \longrightarrow \mathrm{Hom}_{A_f}(A_f^n, N_f) \longrightarrow \mathrm{Hom}_{A_f}(A_f^l, N_f)$$

whence

$$\mathrm{Hom}_A(M,N)_f \cong \ker\left(\mathrm{Hom}_{A_f}(A_f^n, N_f) \to \mathrm{Hom}_{A_f}(A_f^l, N_f)\right) \cong \mathrm{Hom}_{A_f}(M_f, N_f).$$

We conclude that $\mathcal{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$ is indeed coherent.

Finally, we show that $\mathscr{E}xt^i(\mathscr{F},\mathscr{G})$ is coherent for all i. The question is local, so we will take X to be affine. Then X is noetherian and $\mathscr{F}$ is coherent, so we can take an exact sequence $0 \to \mathscr{K} \to \mathscr{E} \to \mathscr{F} \to 0$ with $\mathscr{E}$ locally free and $\mathscr{K}$ coherent. Take the corresponding long exact sequence.

$$0 \longrightarrow \mathcal{H}om(\mathscr{F},\mathscr{G}) \longrightarrow \mathcal{H}om(\mathscr{E},\mathscr{G}) \longrightarrow \mathcal{H}om(\mathscr{K},\mathscr{G}) \longrightarrow \cdots$$

$$\cdots \longrightarrow \mathscr{E}xt^i(\mathscr{F},\mathscr{G}) \longrightarrow \mathscr{E}xt^i(\mathscr{E},\mathscr{G}) \longrightarrow \mathscr{E}xt^i(\mathscr{K},\mathscr{G}) \longrightarrow \cdots$$

(6.7) and (6.3) give us that

$$\mathscr{E}xt^i(\mathscr{E},\mathscr{G}) \cong \mathscr{E}xt^i(\mathscr{O}_X, \mathscr{E}^\vee \otimes \mathscr{G}) = 0$$

for all $i > 0$, we have

$$\mathscr{E}xt^1(\mathscr{F},\mathscr{G}) \cong \mathrm{cok}(\mathcal{H}om(\mathscr{E},\mathscr{G}) \to \mathcal{H}om(\mathscr{K},\mathscr{G})),$$
$$\mathscr{E}xt^i(\mathscr{F},\mathscr{G}) \cong \mathscr{E}xt^{i-1}(\mathscr{K},\mathscr{G}) \ \ \forall\, i \geqslant 2.$$

$\mathscr{K}$ is coherent, so the result holds by induction on i.

(b) $\mathscr{G}$ coherent was only used to show that $\mathrm{Hom}_A(M,N)$ is finitely generated, so we're done. $\square$